Fundamentals of enzyme kinetics and thermodynamic analysis of the metabolic networks for microbial communities

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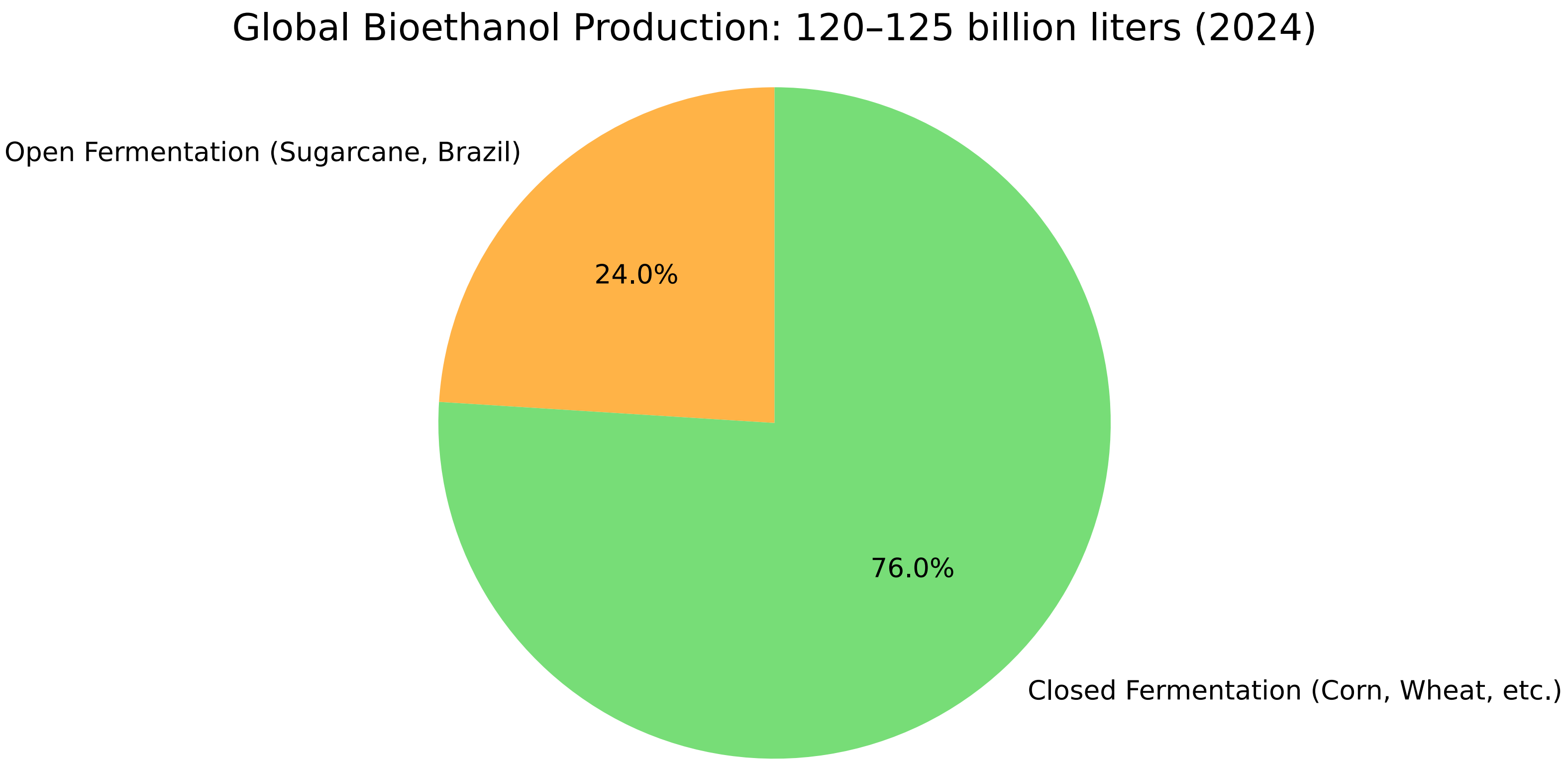
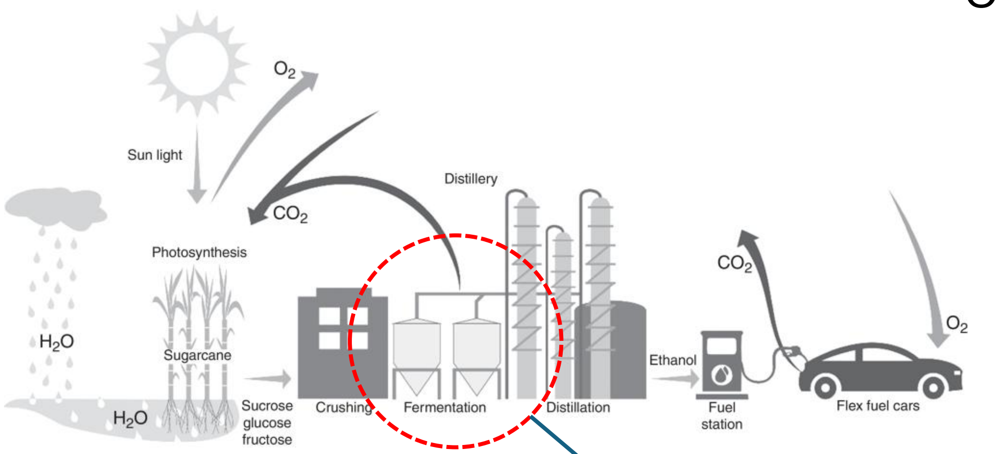
The size of the problem we are going to study today

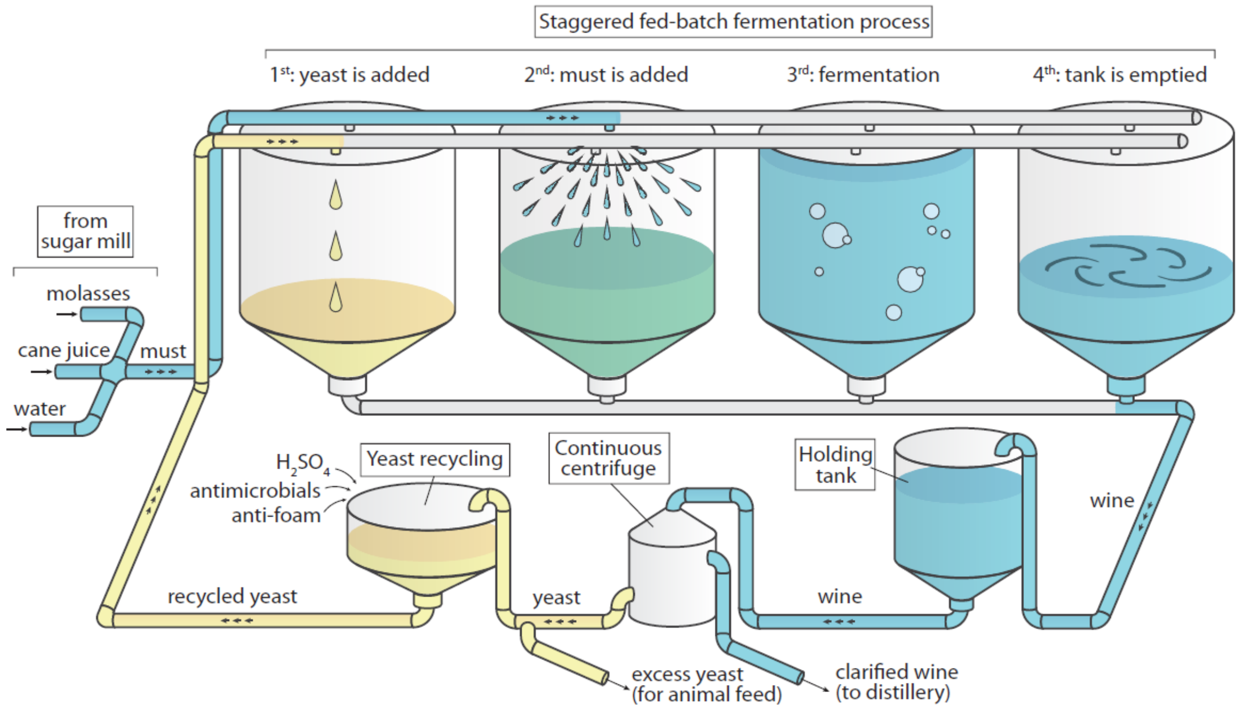


Open fermentation



Source: https://doi.org/10.1016/j.bjm.2016.10.003

(a) Bioethanol production with yeast recycling



Source: https://doi.org/10.1093/g3journal/jkad104

Challenges of the open fermentations

OXFORD
G3 Genes | Genomes | Genetics
G3, 2023, 13(7), jkad104
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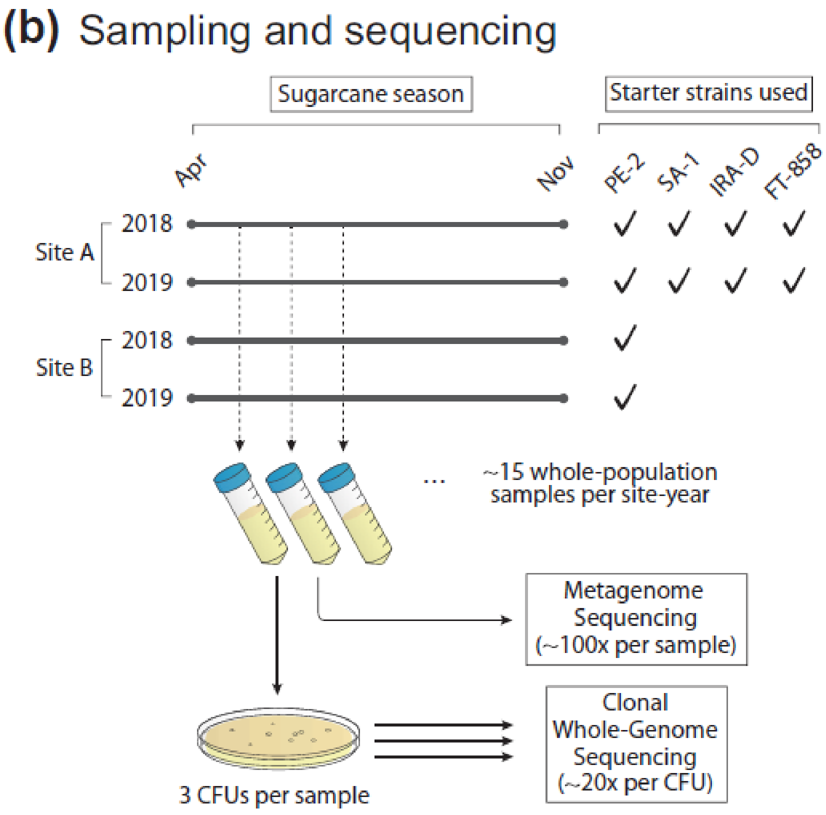
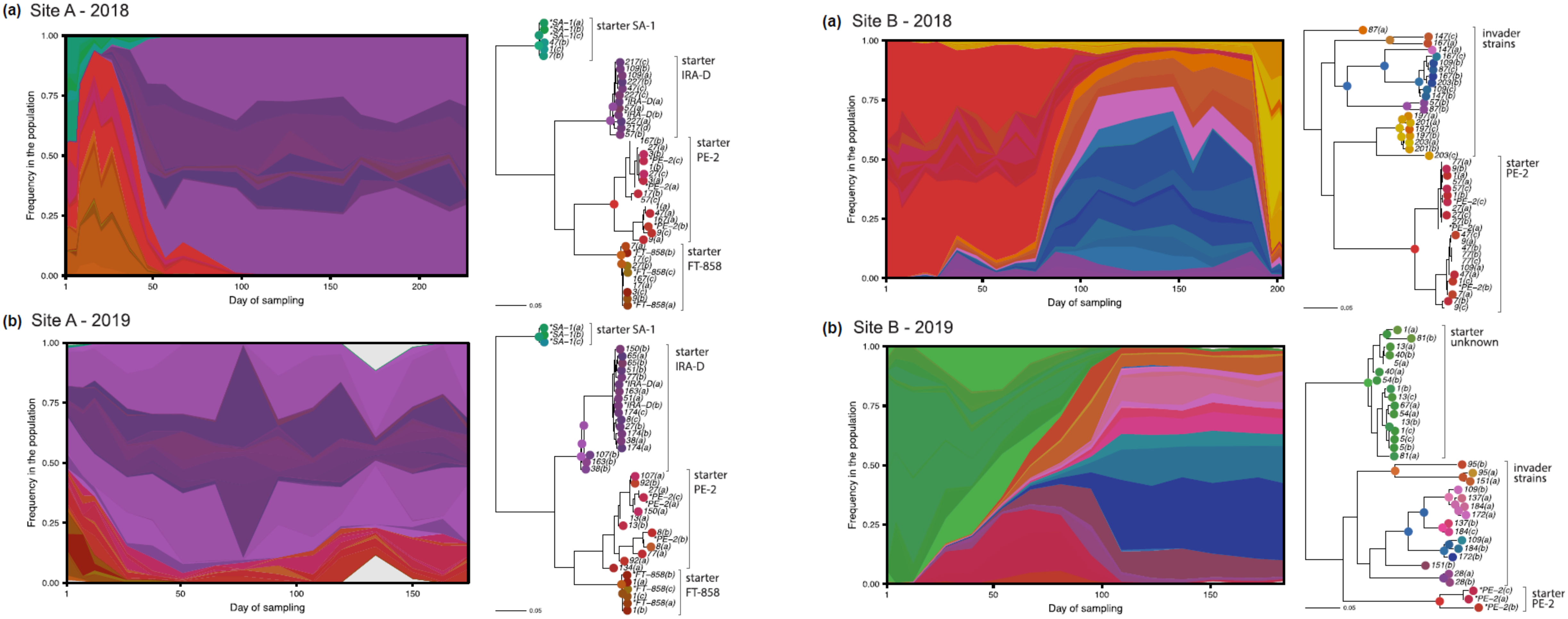
Advance Access Publication Date: 2 June 2023
Investigation

Yeast population dynamics in Brazilian bioethanol production

Artur Rego-Costa,[1,†] I-Ting Huang,[1,†] Michael M. Desai,[1,2,3,4] Andreas K. Gombert[5,*]



The yeast populations are dynamic



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Strain dynamics of contaminating bacteria modulate the yield of ethanol biorefineries

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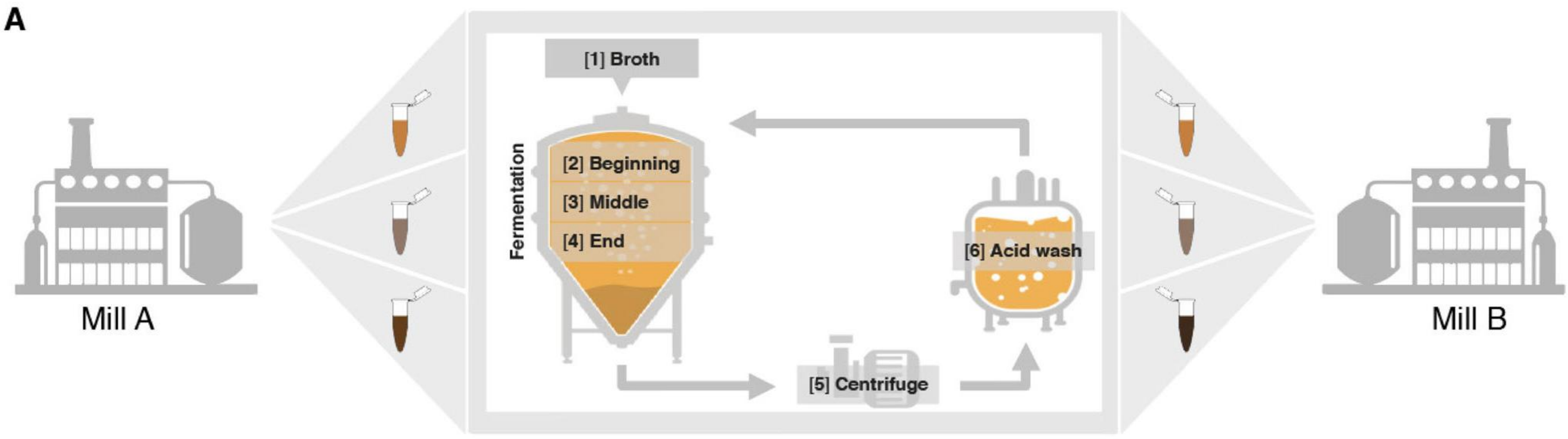
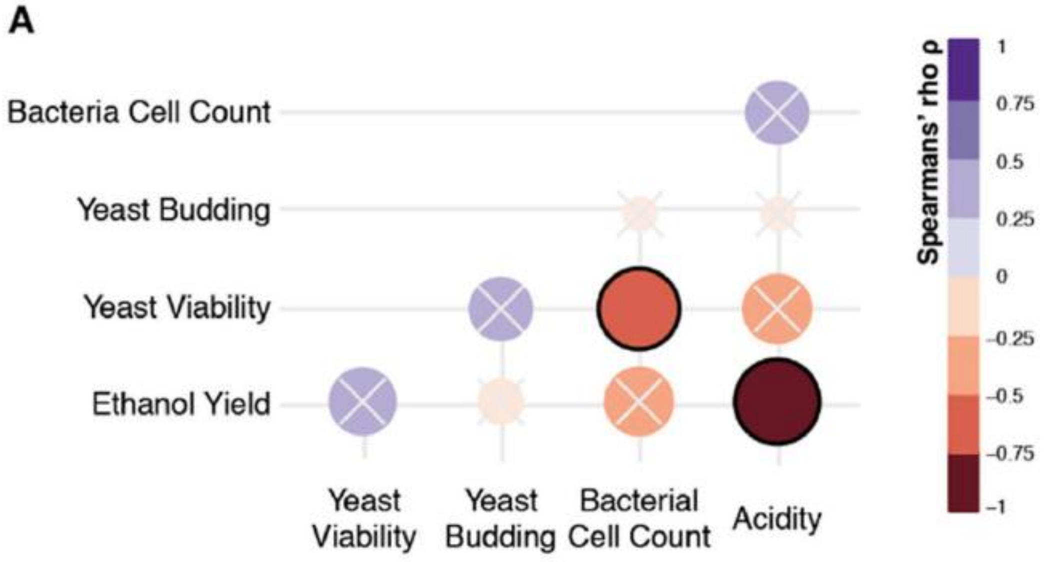
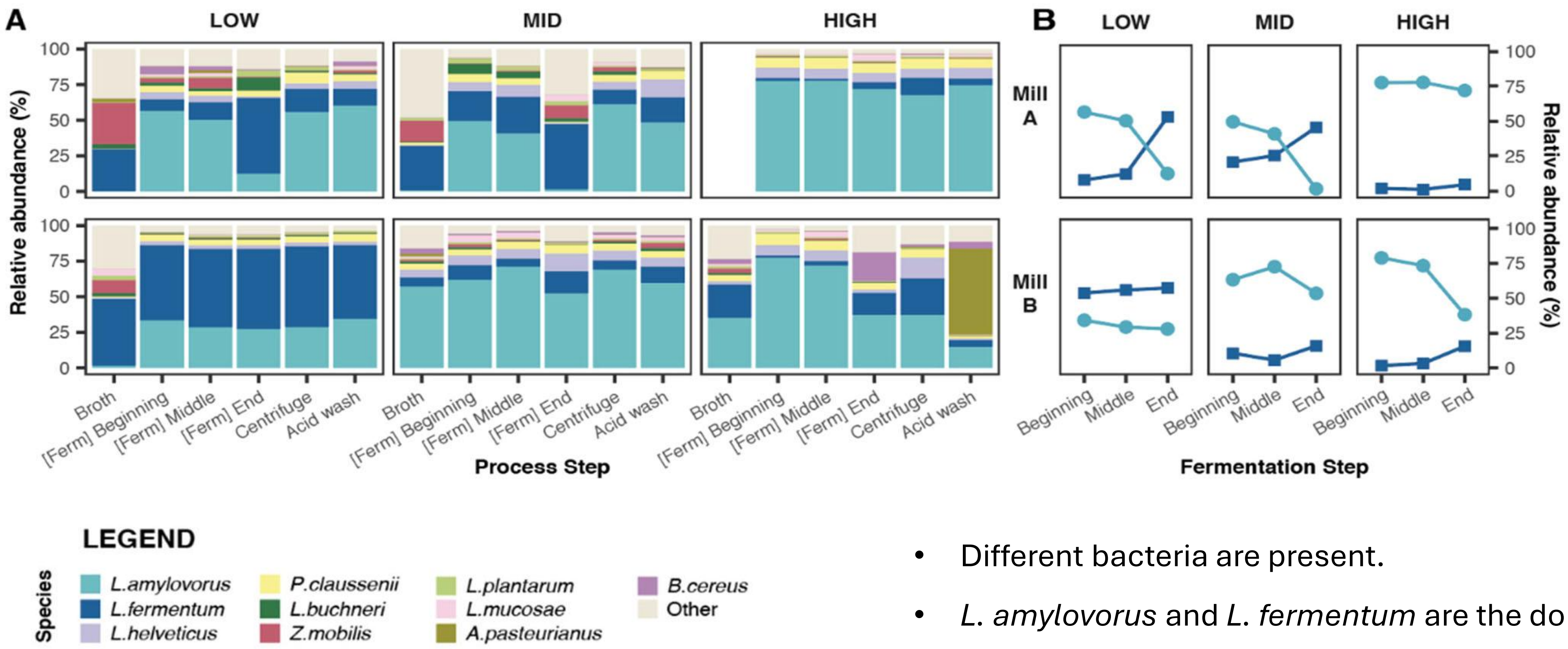
Felipe Senne de Oliveira Lino[1,6], Shilpa Garg[1,6], Simone S. Li[1,2], Maria-Anna Misiakou[1], Kang Kang[3], Bruno Labate Vale da Costa[4], Tobias Svend-Aage Beyer-Pedersen[1], Thamiris Guerra Giacon[5], Thiago Olitta Basso[5], Gianni Panagiotou[3] & Morten Otto Alexander Sommer[1]



Number of bacteria and the acids produced by them negatively affect ethanol production





- Different bacteria are present.
- *L. amylovorus* and *L. fermentum* are the dominant bacteria
- High-performing batches contained more *L. amylovorus*.
- Across all batches and mills, *L. fermentum* increases and *L. amylovorus* decreases by the end of fermentation.
- High-performing batches showed increases in *L. amylovorus* while other lactic acid bacteria including *L. fermentum*, *L. buchneri* and *L. plantarum* decreased.

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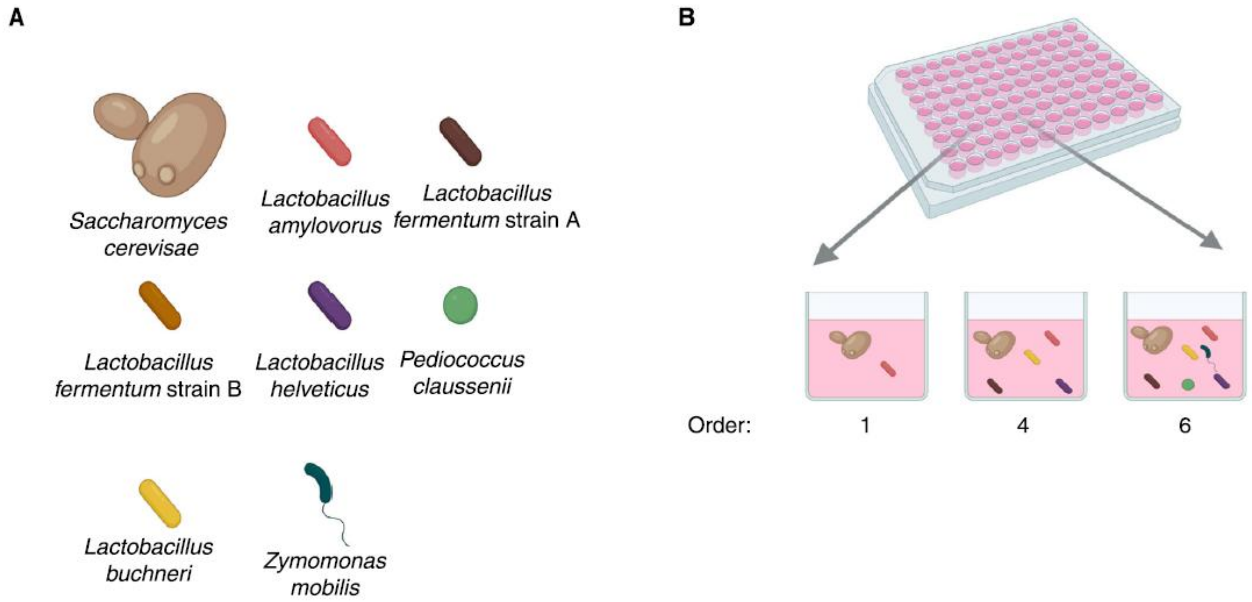
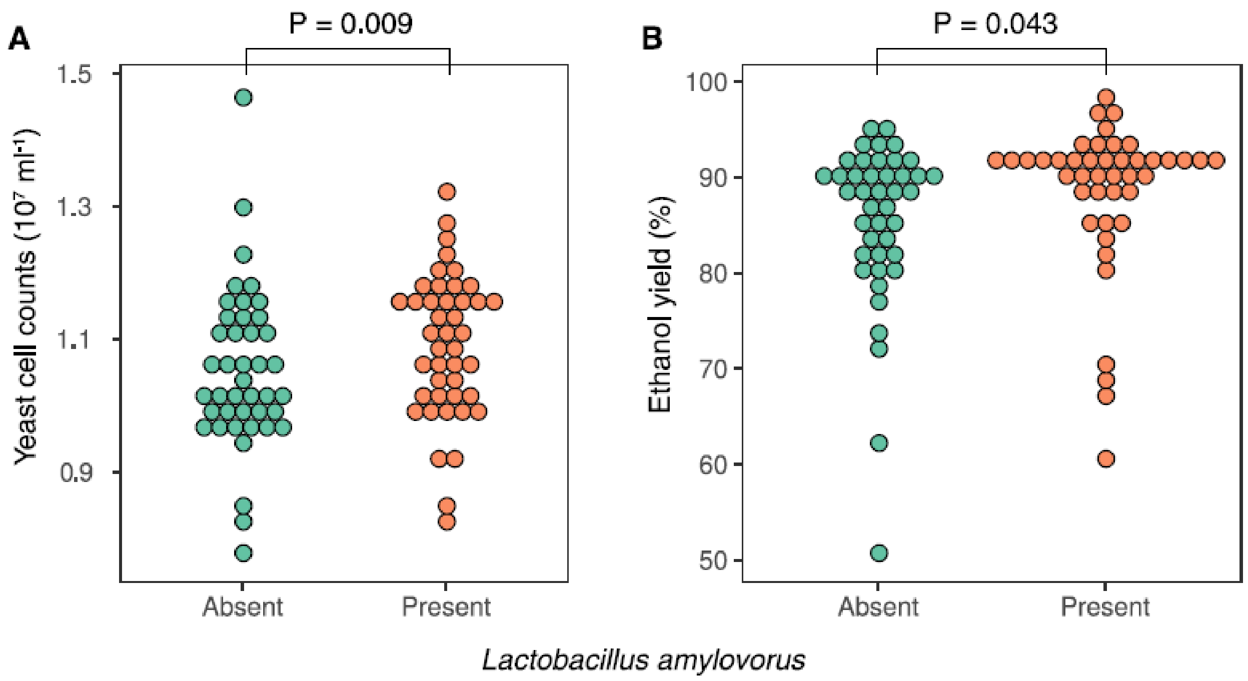
Complex yeast–bacteria interactions affect the yield of industrial ethanol fermentation

Felipe Senne de Oliveira Lino, Djordje Bajic, Jean Celestin Charles Vila, Alvaro Sánchez & Morten Otto Alexander Sommer

Nature Communications **12**, Article number: 1498 (2021) | Cite this article



The bacterium *Lactobacillus amylovorus* has a positive effect on the ethanol production



In open fermentations:

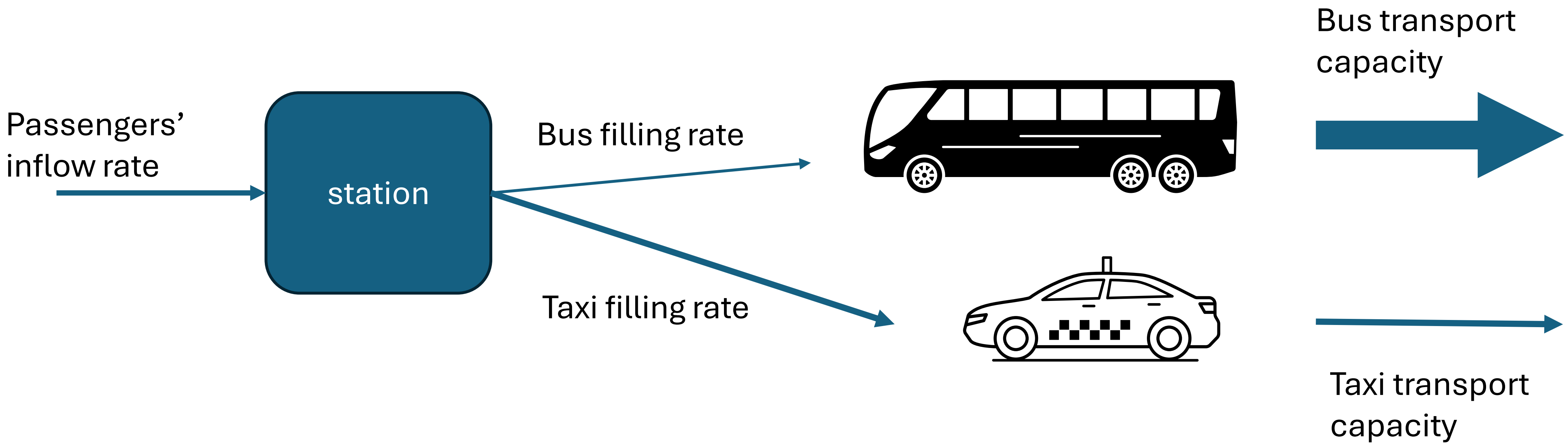
- Yeast populations are dynamic
- Different bacteria are present and influence the ethanol production
- Some bacteria have negative effects on the ethanol production, but some bacteria have a positive effect.

Our goals:

- Concepts from enzyme kinetics are useful to study this problem (first lecture)
- Showing the abyss of the analysis of metabolic networks (first lecture)
- One way to go out of the abyss (Timmy’s lecture)
- Thermodynamics also give us bridges to cross that abyss (second lecture)
- Integrating enzyme kinetics and thermodynamics (second lecture)

Let's start with a fundamental concept: saturation

- One bus station
- Passengers arriving to the station at different rates
- Passengers can be transported out of the station by bus or by taxi



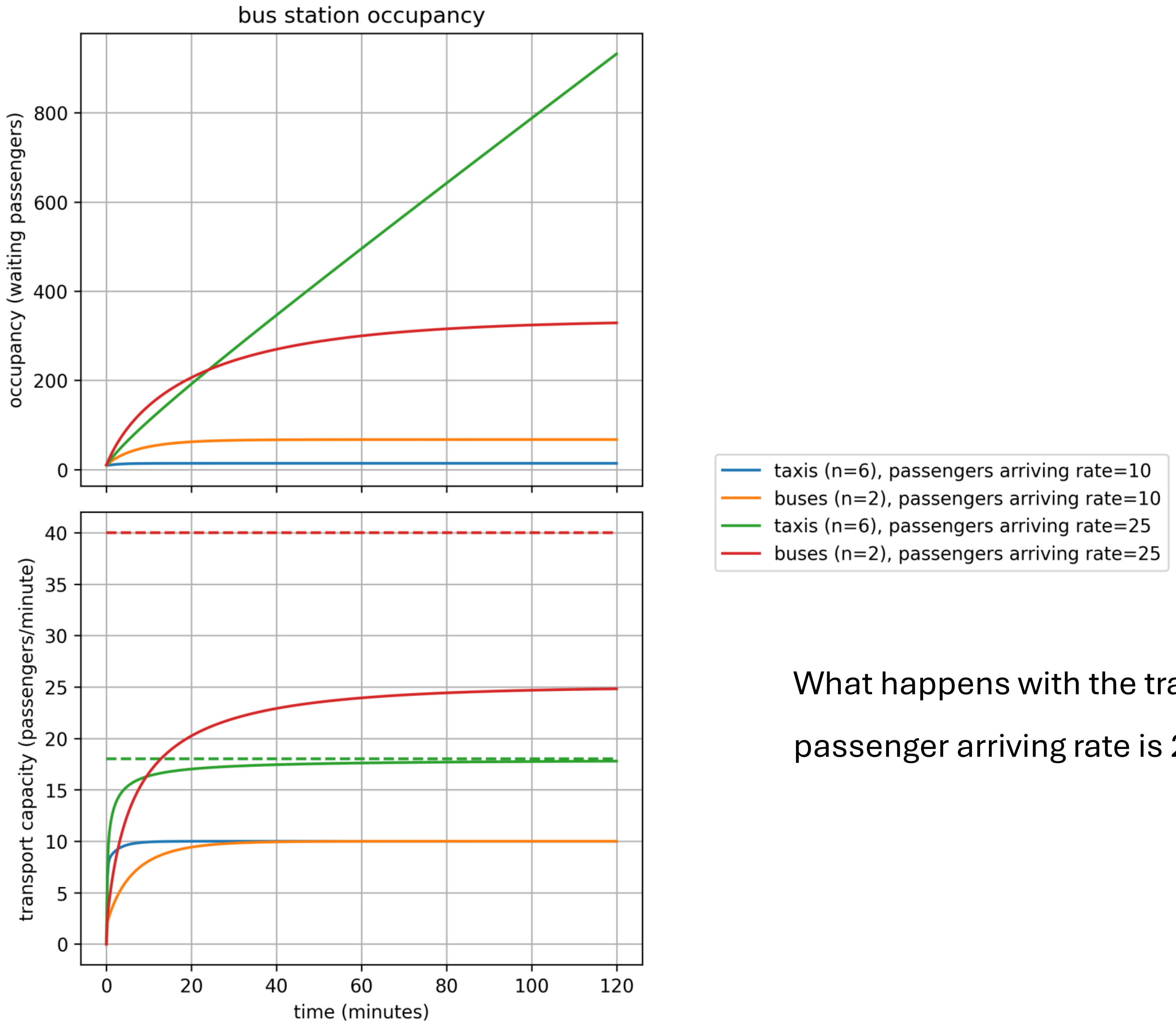
Which is the most effective way of transportation?

S : passengers waiting at the station
E : empty vehicles ready to be filled
ES : vehicles currently being filled with passengers
P : cumulative passengers already transported

k1 : rate at which passengers enter vehicles
k1r : rate at which passengers leave vehicles (before departure)
k2 : rate at which filled vehicles depart
f : external passenger influx (new arrivals per minute)

```
base_scenarios = {
    "taxis": (0.3, 0.3, 3, n_taxis),    # taxis are easy to fill but small capacity, number of taxis
    "buses": (0.1, 0.1, 20, n_buses)    # buses are harder to fill but large capacity, number of buses
}
```

```
dS  = f - k1*S*E + k1r*ES        # passengers at the station
dE  = -k1*S*E + (k1r + k2)*ES    # empty vehicles
dES = k1*S*E - (k1r + k2)*ES     # filled vehicles
dP  = k2*ES                      # passengers transported out of the station
```



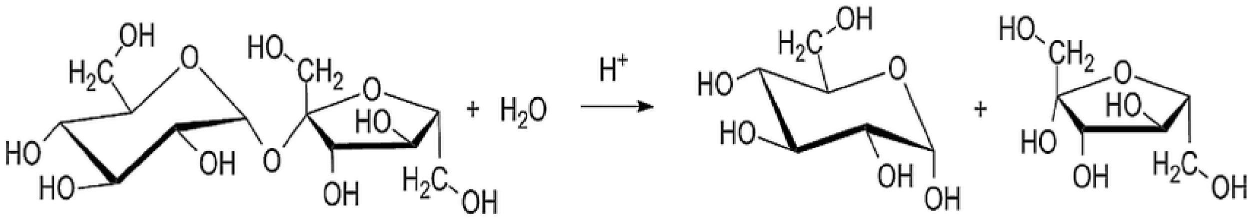
What happens with the transportation by taxi when the passenger arriving rate is 25 passengers/minute?

Maud Menten

Leonor Michaelis

Michaelis, L., and Menten, M. (***1913***) Die kinetik der invertinwirkung, *Biochemistry Zeitung 49, 333-369.*

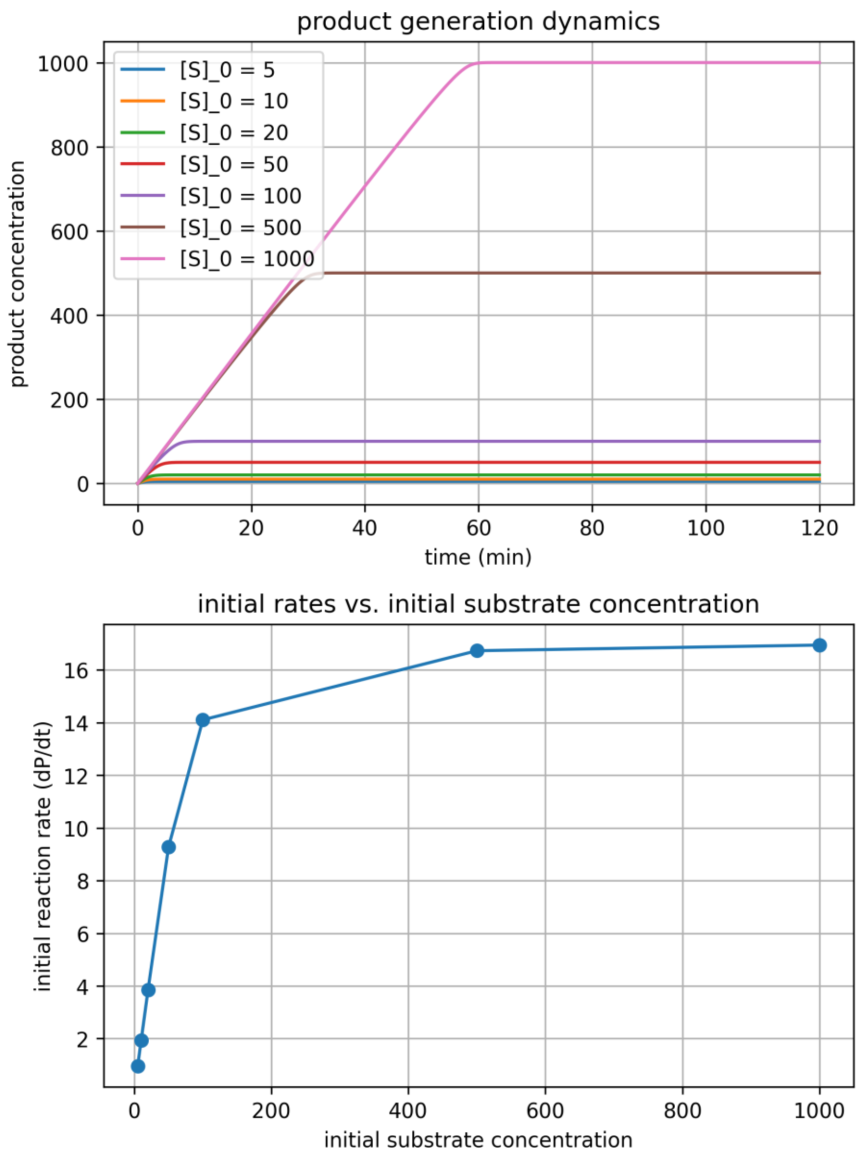
Sucrose hydrolisis by invertase



Sucrose **Glucose** **Fructose**

If there is one liter sucrose solution of 5 grams/liter, and we add 0.001 grams of invertase from yeast, how long it takes to hydrolyze half of the sucrose?

What do we need to know to answer this question?


product generation dynamics
[S]_0 = 5
[S]_0 = 10
[S]_0 = 20
[S]_0 = 50
[S]_0 = 100
[S]_0 = 500
[S]_0 = 1000
product concentration
time (min)
initial rates vs. initial substrate concentration
initial reaction rate (dP/dt)
initial substrate concentration

Goal: predict the product formation rate in an enzyme-catalyzed reaction

$$E + S \underset{k_{1r}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

E: concentration of free enzyme

S: concentration of sucrose

ES: concentration of the enzyme-substrate complex

They assumed that a ***rapid equilibrium*** between the species E, S and ES is established:

$k_1 >> k_2$ $k_{1r} >> k_2$

They assumed that the rate-limiting step is the product formation:

$$product\ formation\ rate = k_2 * ES$$

In 1913, no experimental evidence of the existence of ES.

Rapid equilibrium model

$$E + S \underset{k_{1r}}{\overset{k_1}{\rightleftharpoons}} ES$$

$$rate^{ES\ complex\ formation} = rate^{ES\ complex\ dissociation}$$

$$k_1 * \boldsymbol{E(t)} * S(t) = k_{1r} * \boldsymbol{ES(t)}$$

Difficult to measure

$$K_S = \frac{k_{1r}}{k_1} \qquad K_S = \frac{E(t) * S(t)}{ES(t)} \qquad ES(t) = \frac{E(t) * S(t)}{K_S} = \frac{(E_{added} - ES(t)) * S(t)}{K_S} \qquad ES(t) = \frac{E_{added} * S(t)}{K_S + S(t)}$$

$$\boxed{product\ formation\ rate = k_2 * \frac{E_{added} * S(t)}{K_S + S(t)} = \frac{k_2 * E_{added} * S(t)}{K_S + S(t)} = \frac{V^{max} * S(t)}{K_M + S(t)}}$$

If there is one liter sucrose solution of 5 grams/liter, and we add 0.001 grams of invertase from yeast, **how long** it takes to hydrolyze **half** of the sucrose?

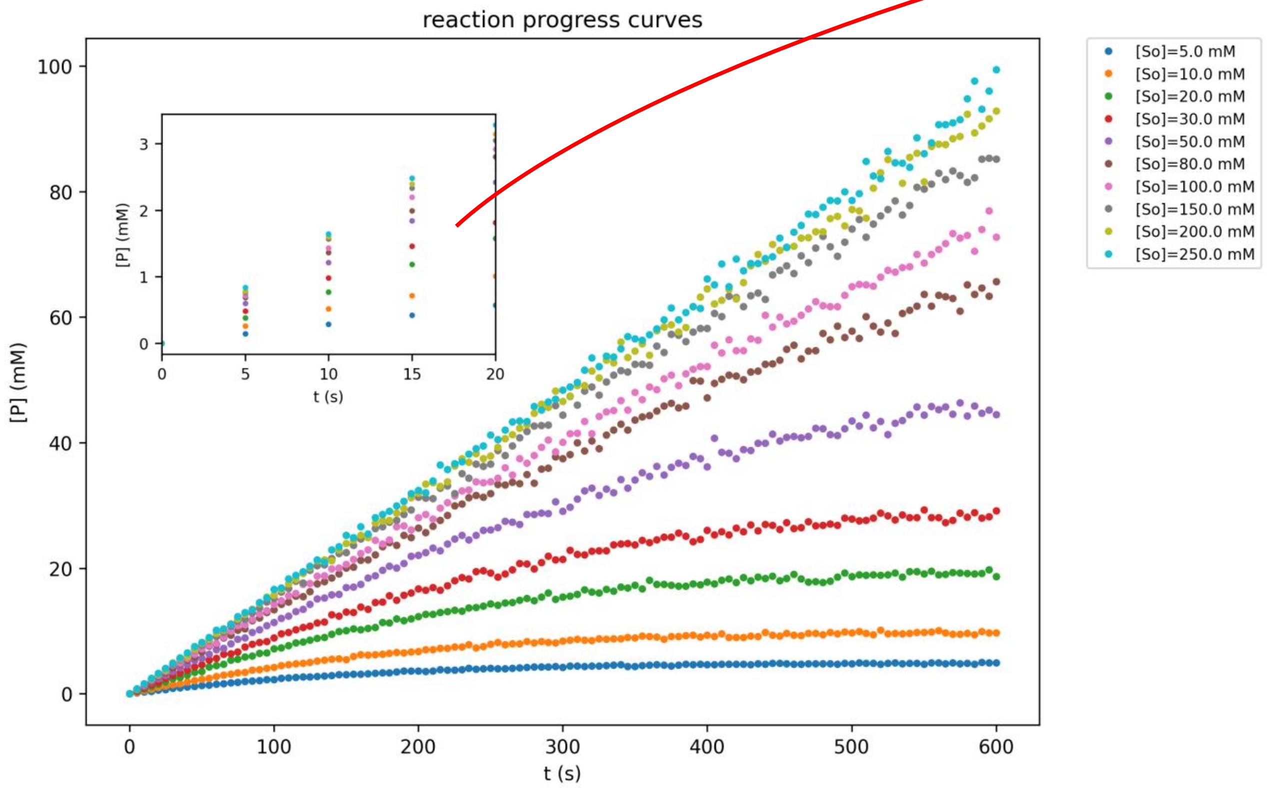
$$product\ formation\ rate = k_{cat} * \frac{E_{added} * S(t)}{K_M + S(t)} = \frac{k_{cat} * E_{added} * S(t)}{K_M + S(t)}$$

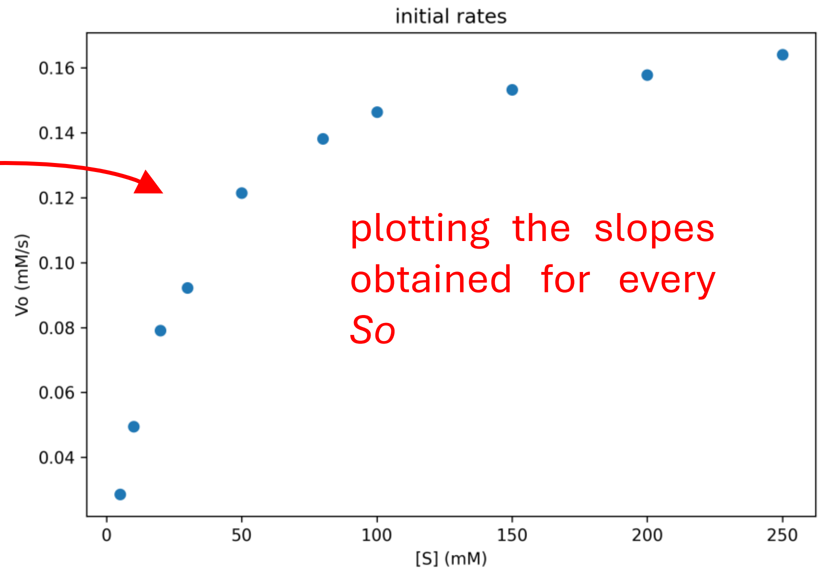
We know *S* at time zero (*So*=5 g/L), and we know E_{added} (0.001 g/L). However,

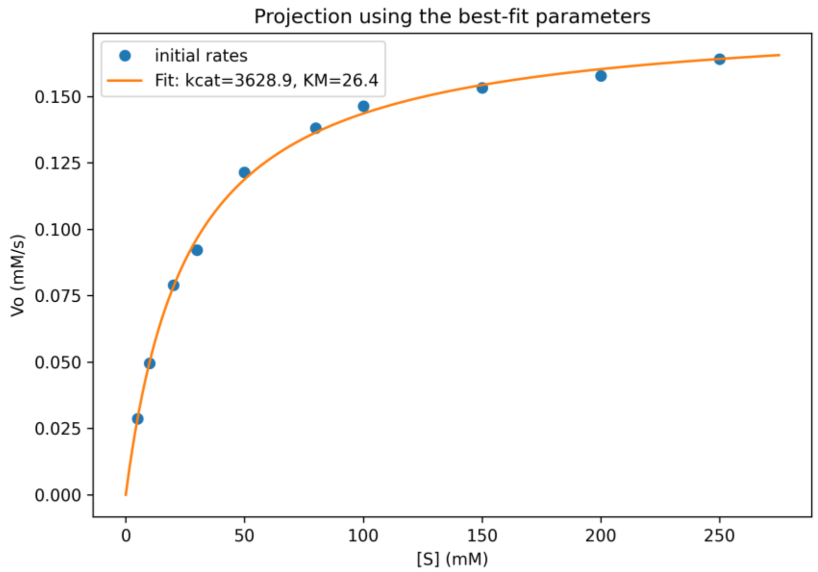
- How can we determine the kinetic parameters k_{cat} and K_M?
- S is changing in time.
- Where is the variable ***time***?

How can we determine the kinetic parameters k_{cat} and K_M?

At the beginning of the reaction, *S* is constant $\longrightarrow$ $$initial\ rate\ (V_o) = \frac{k_{cat} * E_{added} * So}{K_M + So}$$







- S is changing in time. When half of the sucrose has been consumed, we are far from the initial rate conditions
- Where is the variable ***<u>time</u>***?

$$\cancel{initial\ rate\ (V_o) = \frac{k_{cat} * E_{added} * So}{K_M + So}}$$

$$product\ formation\ rate = \frac{dP}{dt} = -\frac{dS}{dt} = \frac{k_{cat} * E * S(t)}{K_M + S(t)}$$

$$-\frac{dS}{dt} = \frac{k_{cat} * E * S(t)}{K_M + S(t)}$$

$$S(t) + K_M * ln\frac{S(t)}{S_o} = -k_{cat} * E * t + S_o$$

Integrated form of the Michaelis-Menten equation

$$t = \frac{S(t) + K_M * ln\frac{S(t)}{So} - So}{-k_{cat} * E}$$

$$t = \frac{S(t) + K_M * \ln\frac{S(t)}{So} - So}{-k_{cat} * E}$$

In our particular case, *S(t)* = *So*/2

$$t_{0.5} = \frac{So/2 + K_M * \ln\frac{1}{2} - So}{-k_{cat} * E}$$

For *So*=5 g*/L, *E* = 0.001 g*/L, K_M = 25 mM and k_{cat} = 3600 s^{-1}, $t_{0.5}$ ≈ 6 minutes

*Mw_sucrose = 342.3 g/mol
*Mw_invertase = 52000 g/mol

The integrated form of the Michaelis-Menten equation can also be employed to find the values of k_{cat} and K_M using reaction progress curves analysis

$$S(t) + K_M * ln\frac{S(t)}{S_o} = -k_{cat} * E * t + S_o$$

Between 1913 and 1997:

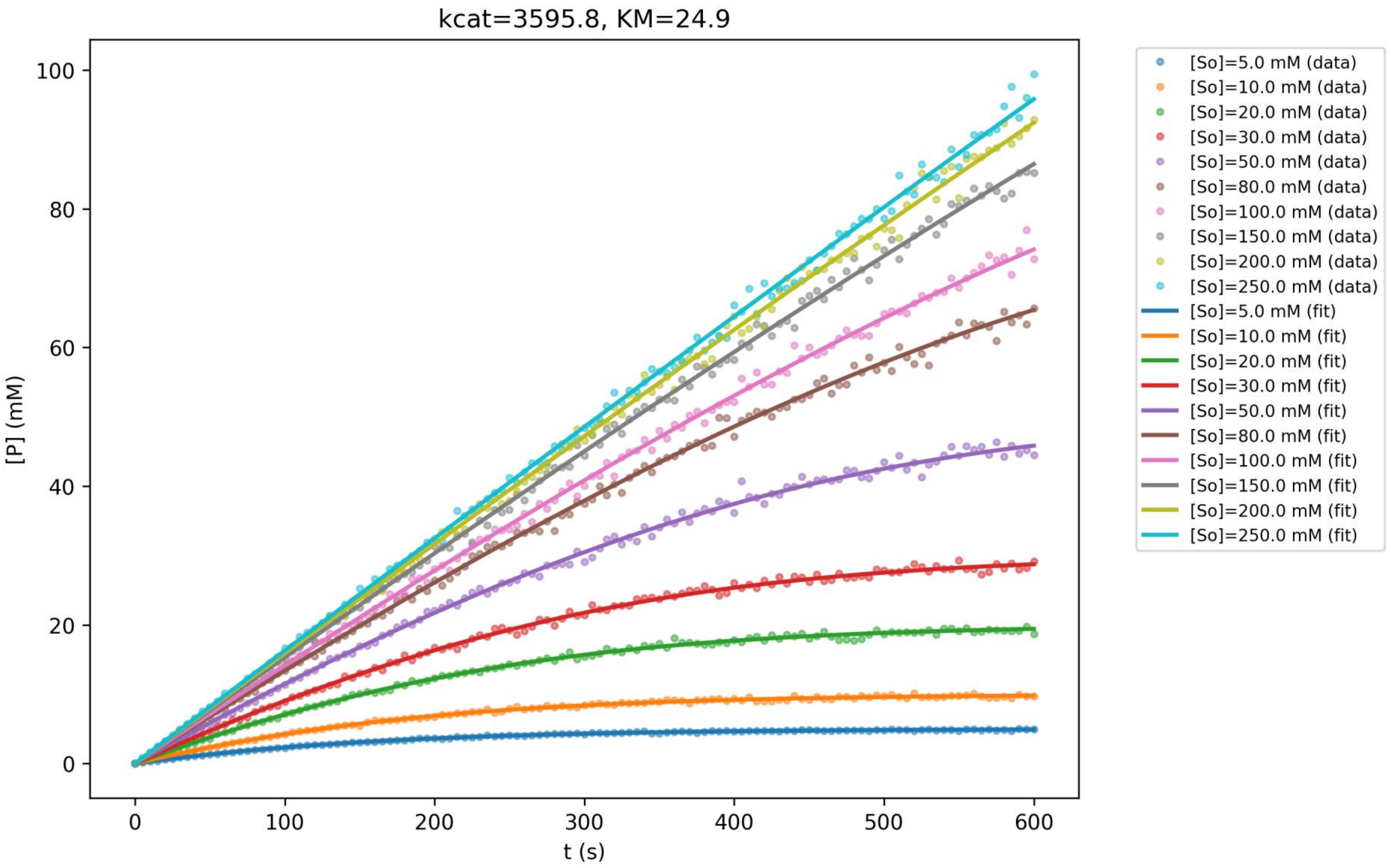
- Numerical integration
- (Complicated) algebraic methods

$$S(t) = K_M * \omega\left(\frac{S_o}{K_M} * e^{\frac{-k_{cat*E*t+S_o}}{K_M}}\right)$$

Latin got talent!

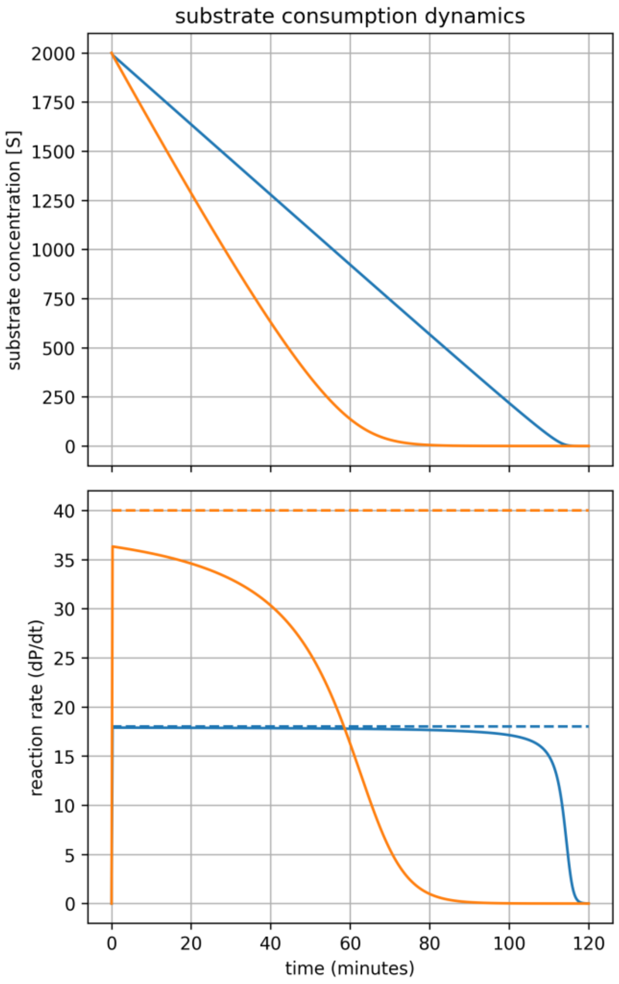
Santiago Schnell Claudio Mendoza

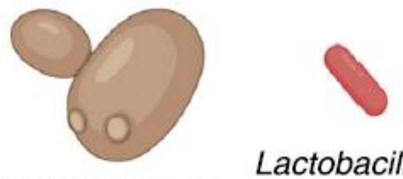
https://doi.org/10.1006/jtbi.1997.0425



kcat=3595.8, KM=24.9
[P] (mM)
t (s)
0
20
40
60
80
100
0
100
200
300
400
500
600
[So]=5.0 mM (data)
[So]=10.0 mM (data)
[So]=20.0 mM (data)
[So]=30.0 mM (data)
[So]=50.0 mM (data)
[So]=80.0 mM (data)
[So]=100.0 mM (data)
[So]=150.0 mM (data)
[So]=200.0 mM (data)
[So]=250.0 mM (data)
[So]=5.0 mM (fit)
[So]=10.0 mM (fit)
[So]=20.0 mM (fit)
[So]=30.0 mM (fit)
[So]=50.0 mM (fit)
[So]=80.0 mM (fit)
[So]=100.0 mM (fit)
[So]=150.0 mM (fit)
[So]=200.0 mM (fit)
[So]=250.0 mM (fit)

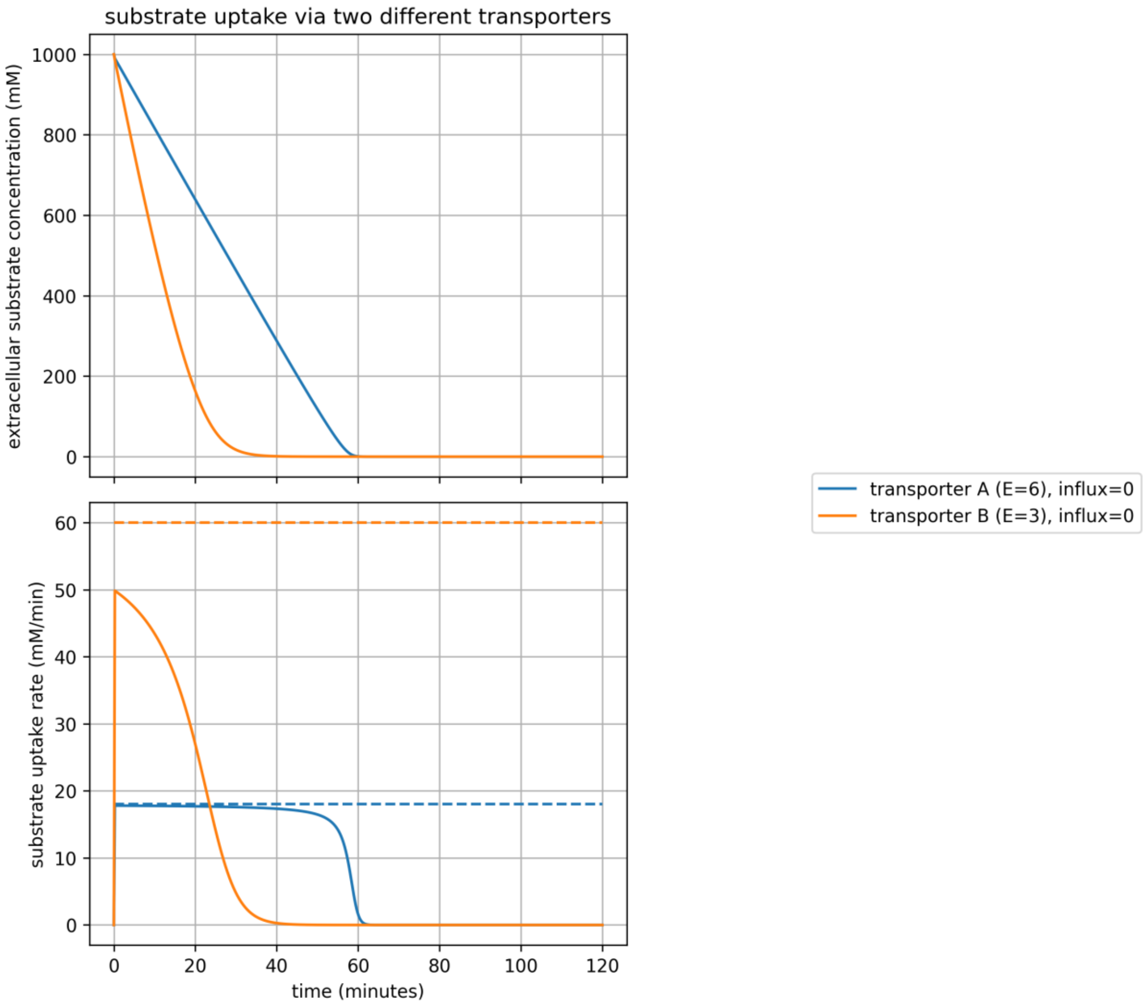
Different enzymes consuming the same substrate and transporters can also be represented with the same equations

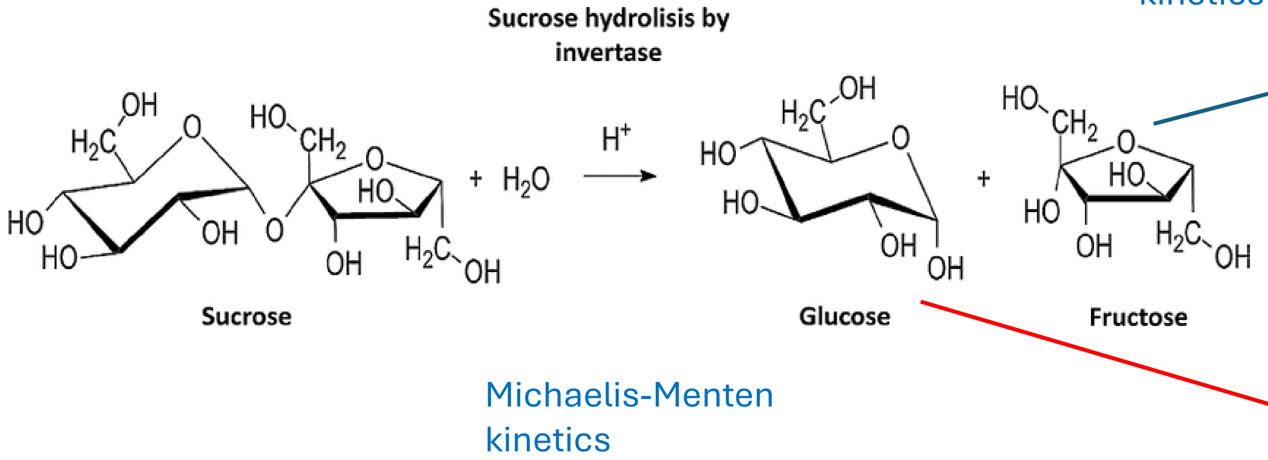


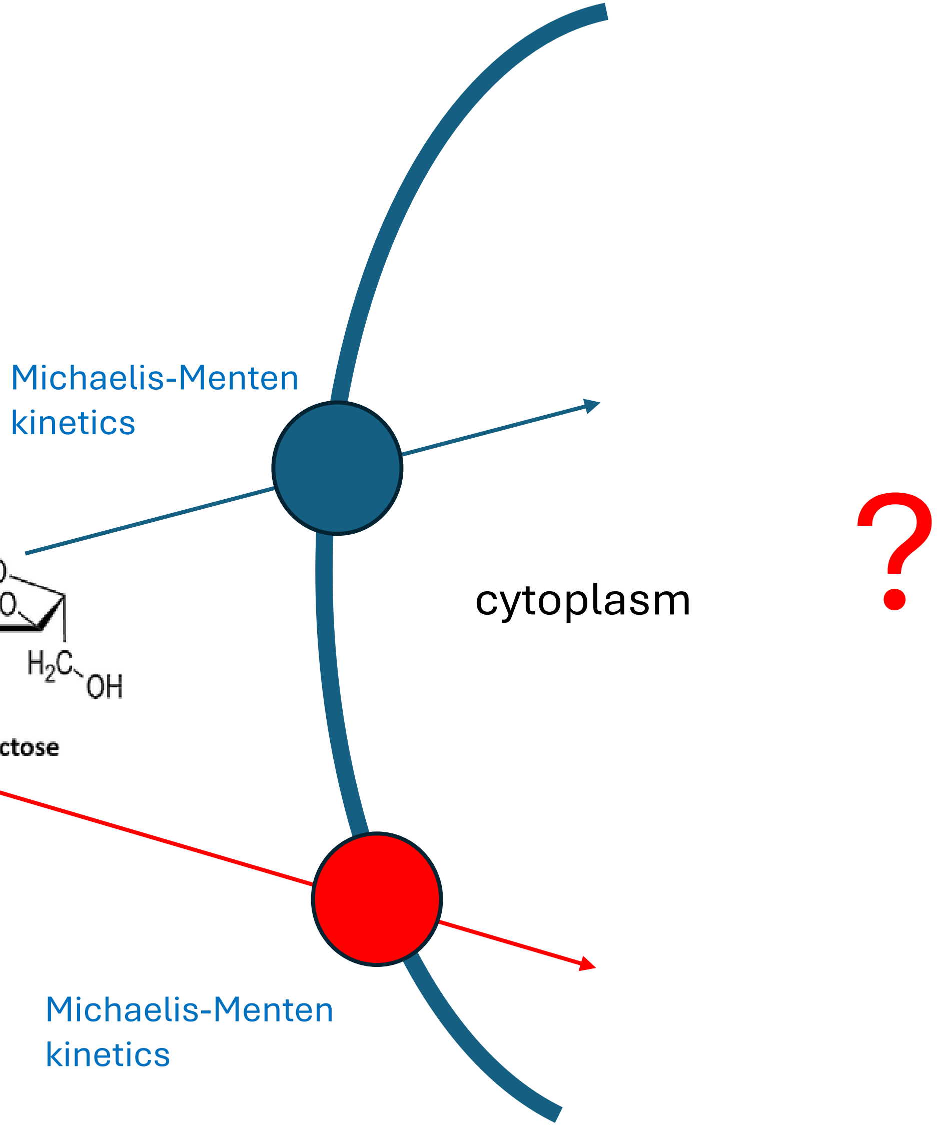


Saccharomyces cerevisae

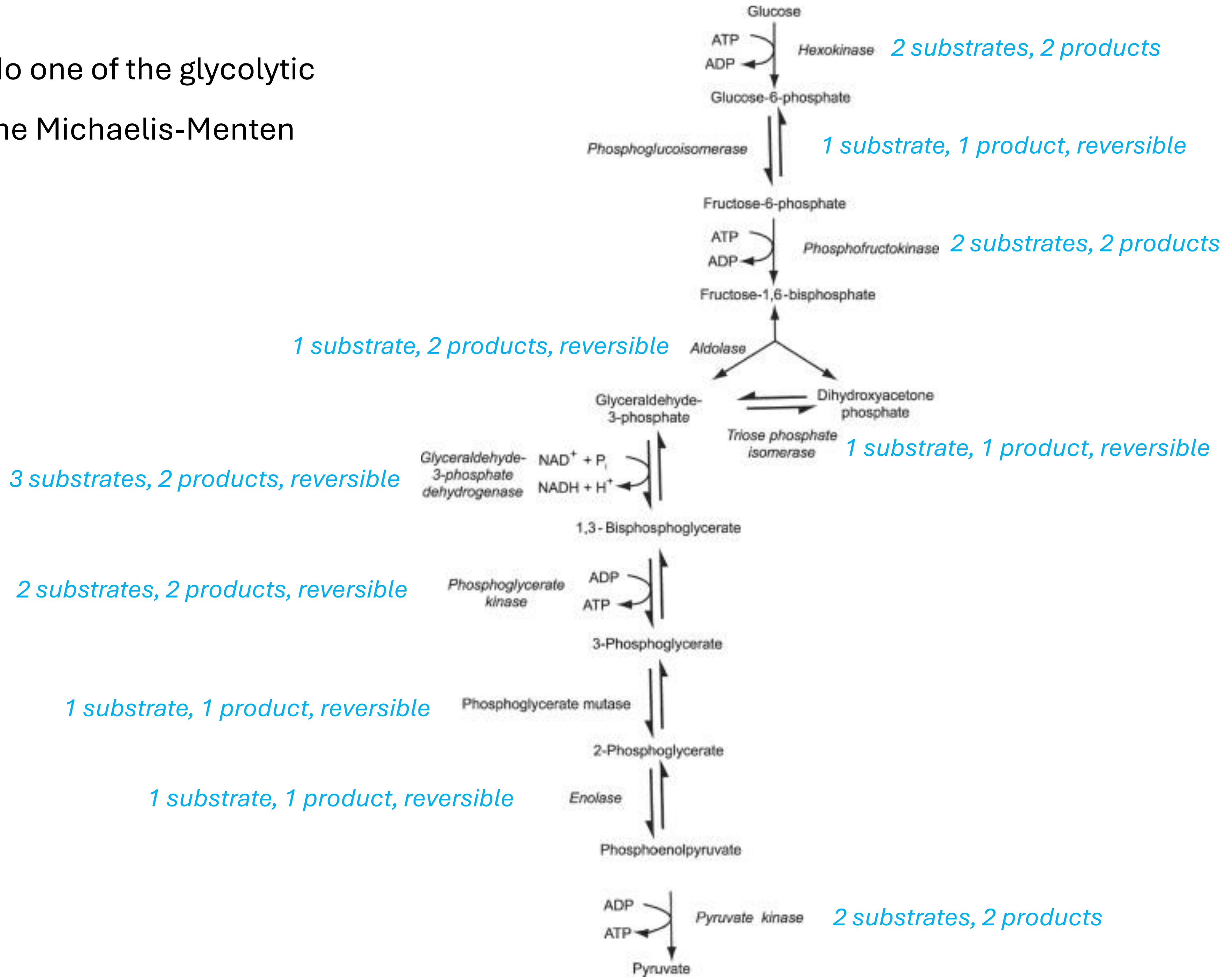
Lactobacillus amylovorus






Sucrose hydrolisis by invertase
H^+
+ H_2O
Sucrose
Glucose
+
Fructose
Michaelis-Menten kinetics
Michaelis-Menten kinetics
Michaelis-Menten kinetics
cytoplasm
?

Houston, we have a problem: No one of the glycolytic reaction can be described with the Michaelis-Menten equation!



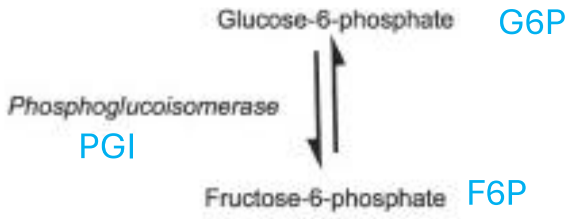
$$E + S \underset{k_{1r}}{\overset{k_1}{\rightleftharpoons}} ES \underset{k_{2r}}{\overset{k_2}{\rightleftharpoons}} E + P$$

Assuming **steady-state** for the enzyme-substrate complex:

$$\frac{dES}{dt} = 0 \qquad k_{cat}^{f} = k_2 \qquad K_M^S = \frac{k_{1r} + k_2}{k_1} \qquad k_{cat}^{r} = k_{1r} \qquad K_M^P = \frac{k_{1r} + k_2}{k_{2r}}$$

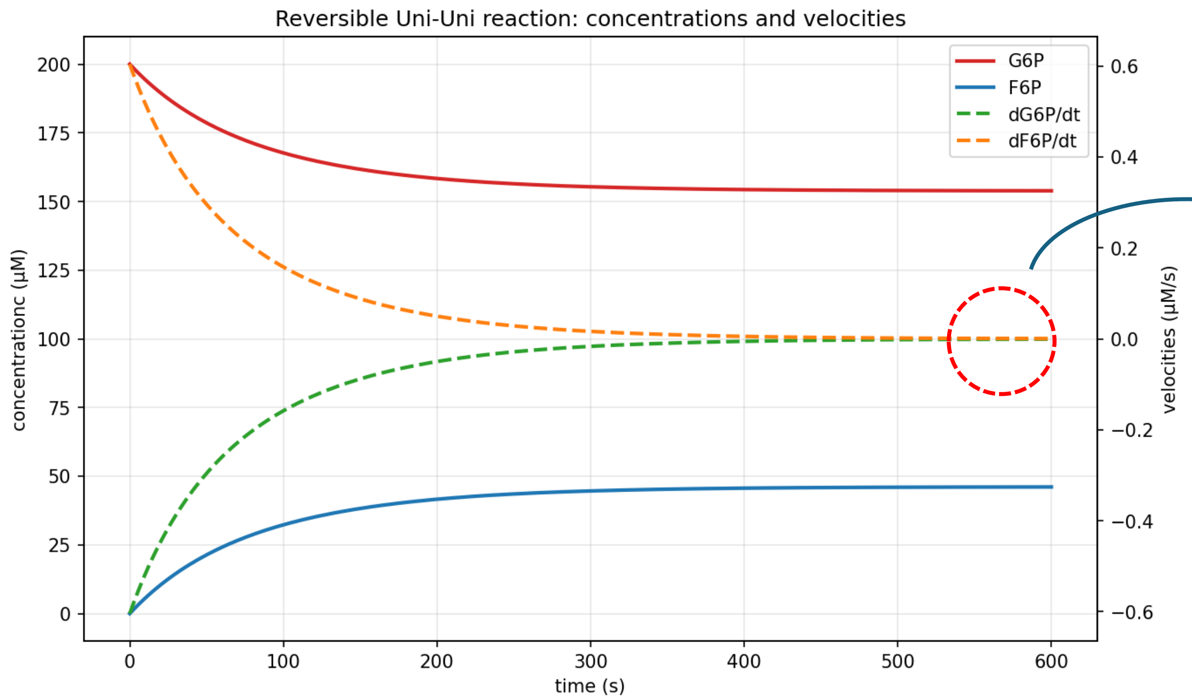
$$\frac{dP}{dt} = -\frac{dS}{dt} = \frac{\frac{k_{cat}^{f}}{K_M^S} * E * S(t) - \frac{k_{cat}^{r}}{K_M^P} * E * P(t)}{1 + \frac{S(t)}{K_M^S} + \frac{P(t)}{K_M^P}}$$

$$\frac{dP}{dt} = -\frac{dS}{dt} = \frac{\frac{k_{cat}^{f}}{K_M^S} * E * S(t) - \frac{k_{cat}^{r}}{K_M^P} * E * P(t)}{1 + \frac{S(t)}{K_M^S} + \frac{P(t)}{K_M^P}}$$



$$\frac{dF6P}{dt} = -\frac{dG6P}{dt} = \frac{\frac{k_{cat}^{f,PGI}}{K_M^{G6P,PGI}} * \mathrm{PGI} * G6P(t) - \frac{k_{cat}^{r,PGI}}{K_M^{F6P,PGI}} * \mathrm{PGI} * F6P(t)}{1 + \frac{G6P(t)}{K_M^{G6P,PGI}} + \frac{F6P(t)}{K_M^{F6P,PGI}}}$$

Let’s imagine that the reaction goes until approach the thermodynamic equilibrium:



$$\frac{dP}{dt} = -\frac{dS}{dt} = 0 = \frac{\frac{k_{cat}^{f}}{K_M^S} * E * S^{eq} - \frac{k_{cat}^{r}}{K_M^P} * E * P^{eq}}{1 + \frac{S^{eq}}{K_M^S} + \frac{P^{eq}}{K_M^P}}$$

$$\frac{k_{cat}^{f}}{K_M^S} * \cancel{E} * S^{eq} = \frac{k_{cat}^{r}}{K_M^P} * \cancel{E} * P^{eq}$$

Haldane’s relationship $$\frac{P^{eq}}{S^{eq}} = K_{eq} = \frac{k_{cat}^{f} * K_M^P}{k_{cat}^{r} * K_M^S}$$

Main considerations regarding the Haldane relationship:

$$K_{eq} = \frac{k_{cat}^{f} * K_M^P}{k_{cat}^{r} * K_M^S}$$

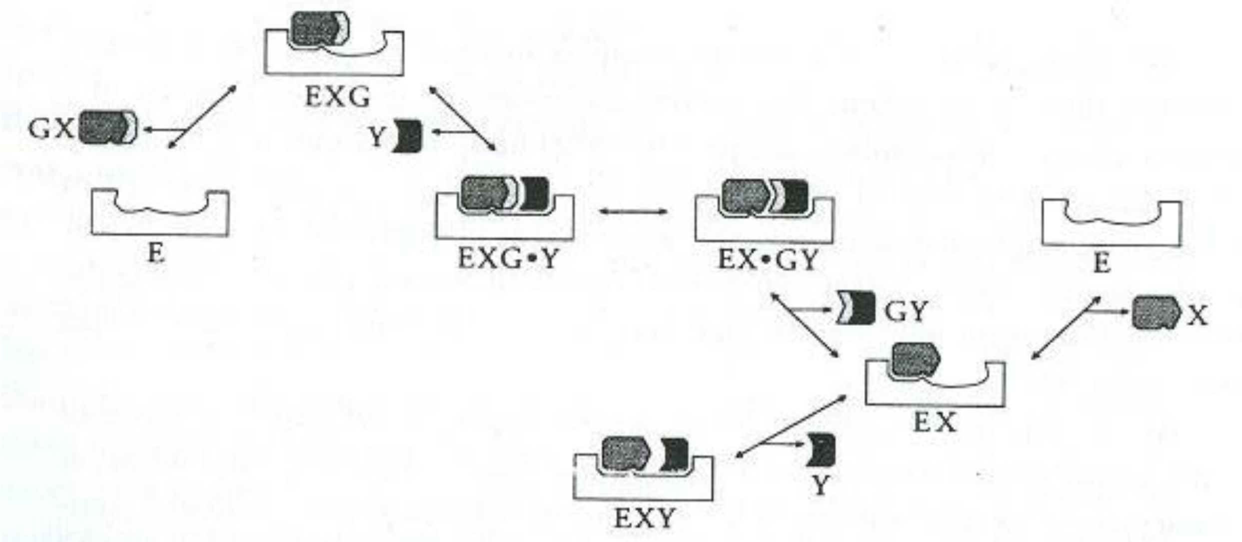
1. the kinetic parameters of the reaction in the forward direction **are not independent** of the kinetic parameters of the reaction in the backward direction.
2. More complex enzyme-catalyzed reactions also have their corresponding (more complex) Haldane relationships.

Multi-substrate reactions

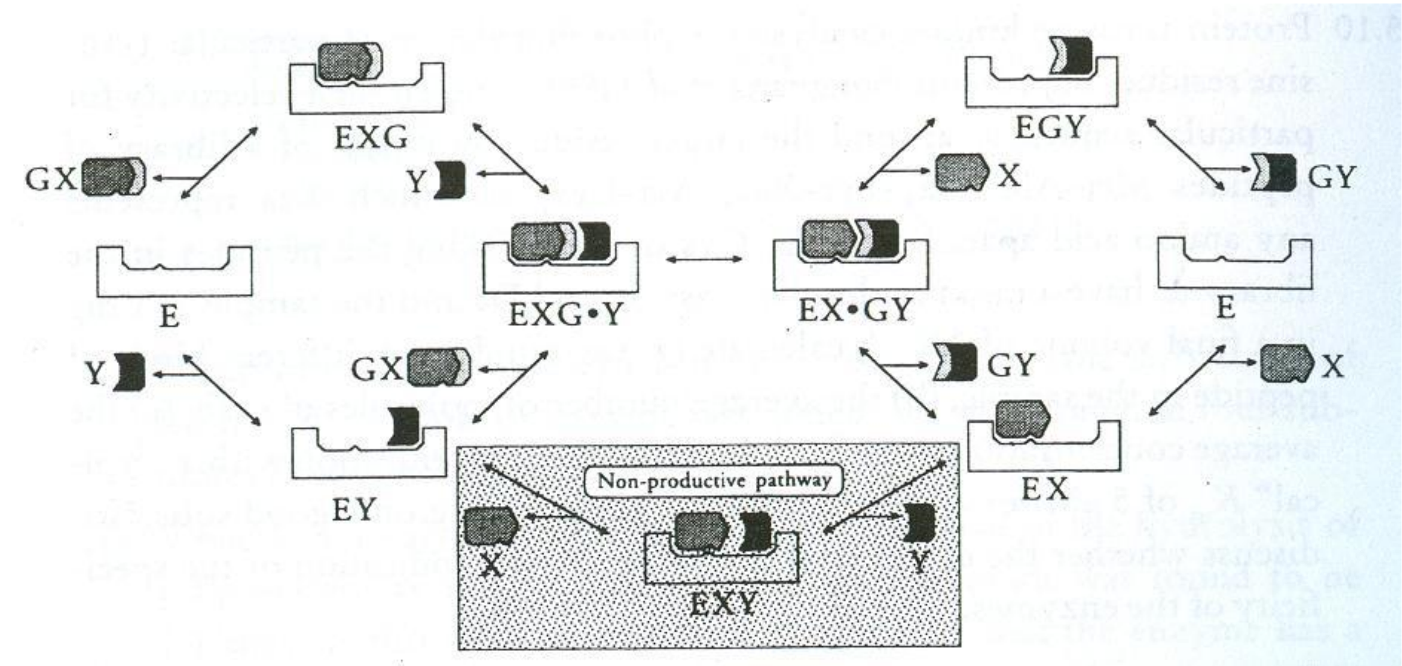
Bi-substrate reactions mechanisms can be classified as follows:

ping-pong mechanism
random substrate binding mechanism
ordered substrate binding mechanism

Example of an **ordered mechanism**



Example of a **random mechanism**



$$v = \frac{\dfrac{V_{+}ab}{K_{iA}K_{mB}} - \dfrac{V_{-}pq}{K_{mP}K_{iQ}}}{1 + \dfrac{a}{K_{iA}} + \dfrac{K_{mA}b}{K_{iA}K_{mB}} + \dfrac{K_{mQ}p}{K_{mP}K_{iQ}} + \dfrac{q}{K_{iQ}} + \dfrac{ab}{K_{iA}K_{mB}} + \dfrac{K_{mQ}ap}{K_{iA}K_{mP}K_{iQ}} + \dfrac{K_{mA}bq}{K_{iA}K_{mB}K_{iQ}} + \dfrac{pq}{K_{mP}K_{iQ}} + \dfrac{abp}{K_{iA}K_{mB}K_{iP}} + \dfrac{bpq}{K_{iB}K_{mP}K_{iQ}}}$$

Eq. 6.5 Fundamentals of Enzyme Kinetics, Athel Cornish-Bowden

How can we obtain the kinetic parameters in this case?

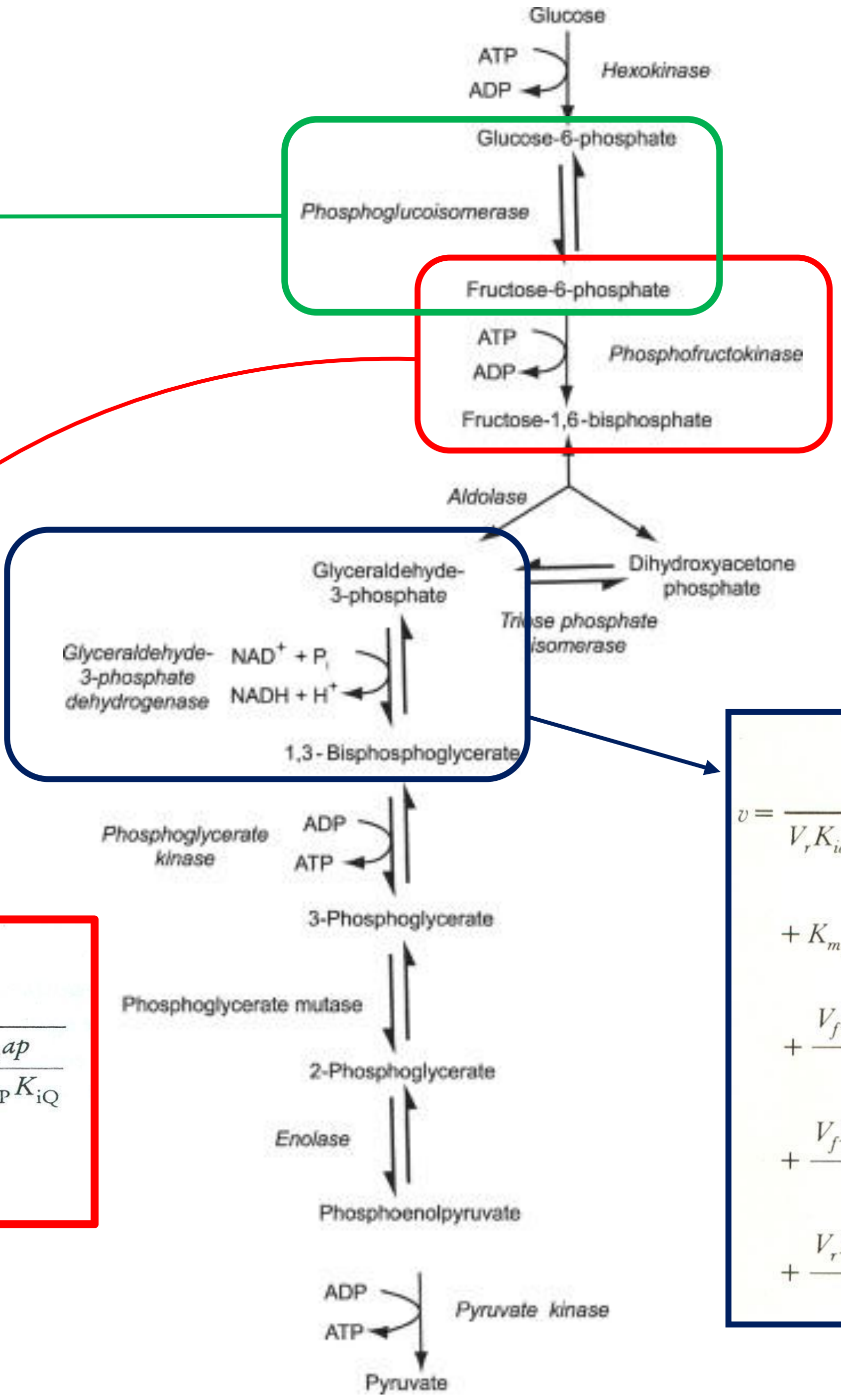
If products (p and q) are at a very low concentration (near zero)

$$v = \frac{\dfrac{V_{+}ab}{K_{iA}K_{mB}} - \cancel{\dfrac{V_{-}pq}{K_{mP}K_{iQ}}}}{1 + \dfrac{a}{K_{iA}} + \dfrac{K_{mA}b}{K_{iA}K_{mB}} + \cancel{\dfrac{K_{mQ}p}{K_{mP}K_{iQ}}} + \cancel{\dfrac{q}{K_{iQ}}} + \dfrac{ab}{K_{iA}K_{mB}} + \cancel{\dfrac{K_{mQ}ap}{K_{iA}K_{mP}K_{iQ}}} + \cancel{\dfrac{K_{mA}bq}{K_{iA}K_{mB}K_{iQ}}} + \cancel{\dfrac{pq}{K_{mP}K_{iQ}}} + \cancel{\dfrac{abp}{K_{iA}K_{mB}K_{iP}}} + \cancel{\dfrac{bpq}{K_{iB}K_{mP}K_{iQ}}}}$$

$$v = \frac{Vab}{K_{iA}K_{mB} + K_{mB}a + K_{mA}b + ab}$$

$$v = \frac{\left(\dfrac{Vb}{K_{mB} + b}\right)a}{\left(\dfrac{K_{iA}K_{mB} + K_{mA}b}{K_{mB} + b}\right) + a} = \frac{V^{app}a}{K_{m}^{app} + a}$$

$$v = \frac{\frac{k_{cat}^{f}}{K_s} * E * s - \frac{k_{cat}^{r}}{K_p} * E * p}{1 + \frac{s}{K_s} + \frac{p}{K_p}}$$



$$v = \frac{\frac{V_{+}ab}{K_{iA}K_{mB}} - \frac{V_{-}pq}{K_{mP}K_{iQ}}}{1 + \frac{a}{K_{iA}} + \frac{K_{mA}b}{K_{iA}K_{mB}} + \frac{K_{mQ}p}{K_{mP}K_{iQ}} + \frac{q}{K_{iQ}} + \frac{ab}{K_{iA}K_{mB}} + \frac{K_{mQ}ap}{K_{iA}K_{mP}K_{iQ}} + \frac{K_{mA}bq}{K_{iA}K_{mB}K_{iQ}} + \frac{pq}{K_{mP}K_{iQ}} + \frac{abp}{K_{iA}K_{mB}K_{iP}} + \frac{bpq}{K_{iB}K_{mP}K_{iQ}}}$$

$$v = \frac{V_f V_r\left([A][B][C] - \frac{[P][Q]}{K_{eq}}\right)}{\begin{array}{l} V_r K_{ia}K_{ib}K_{m_C} + V_r K_{ib}K_{m_C}[A] + V_r K_{ia}K_{m_B}[C] + V_r K_{m_C}[A][B] \\ + K_{m_B}[A][C] + V_r K_{m_A}[B][C] + V_r[A][B][C] + \frac{V_f K_{m_Q}[P]}{K_{eq}} \\ + \frac{V_f K_{m_P}[Q]}{K_{eq}} + \frac{V_f[P][Q]}{K_{eq}} + \frac{V_f K_{m_Q}[A][P]}{K_{ia}K_{eq}} + \frac{V_f K_{m_Q}[A][B][P]}{K_{ia}K_{ib}K_{eq}} \\ + \frac{V_f K_{m_Q}[A][B][C][P]}{K_{ia}K_{ib}K_{ic}K_{eq}} + \frac{V_r K_{ia}K_{m_B}[C][Q]}{K_{iq}} + \frac{V_r K_{m_A}[B][C][Q]}{K_{iq}} \\ + \frac{V_r K_{ia}K_{m_B}[C][P][Q]}{K_{ip}K_{iq}} + \frac{V_r K_{m_A}K_{ic}[B][P][Q]}{K_{ip}K_{iq}} + \frac{V_r K_{m_A}[B][C][P][Q]}{K_{ip}K_{iq}} \end{array}}$$

Fortunately, there are methods to deal with this problem

Analysis of metabolic network under metabolic steady-state

One way to go out of the abyss (Timmy's lecture)

Simplified rate equations and thermodynamic analysis

- Thermodynamics also give us bridges to cross that abyss (second lecture)
- Integrating enzyme kinetics and thermodynamics (second lecture)